

DTU Physics

Sources and Monitors



Sources: In general

- A source component generates Monte Carlo neutrons.
In McStas terms this means:
 - Set the neutron state to something representative of the source we are trying to model.
 - i.e.: insert values in the neutron state vector:
 $\{x, y, z, vx, vy, vz, t, sx, sy, sz, p\}$
drawn from appropriate distributions.
- EXAMPLE:
Neutrons from a uniform wavelength distribution emerging from a circular aperture.

Access the docs



IMPORTANT:

All (and more) of this information can be found in the online pdf component documentation, e.g.

https://github.com/mccode-dev/McCode/raw/main/docpkg/manuals/mcstas/Component_manual.pdf

or

<http://mcstas.org/download/components/doc/manuals/mcstas-components.pdf>

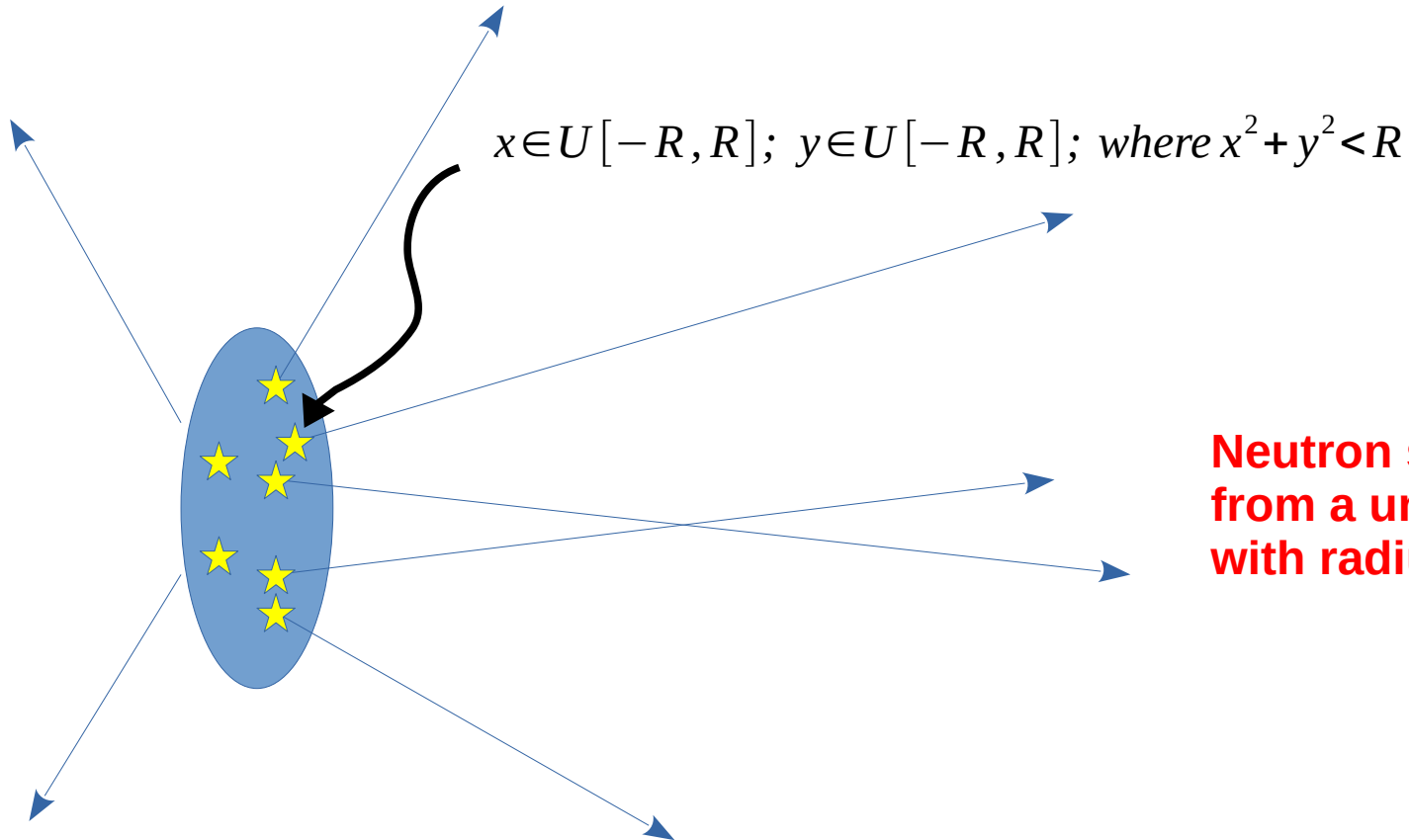
- also distributed with your McStas installation - `mcdoc -c`

The component documentation along with the command:

`"mcdoc <component_you_are_searching_for>"`

are your best friends when using McStas

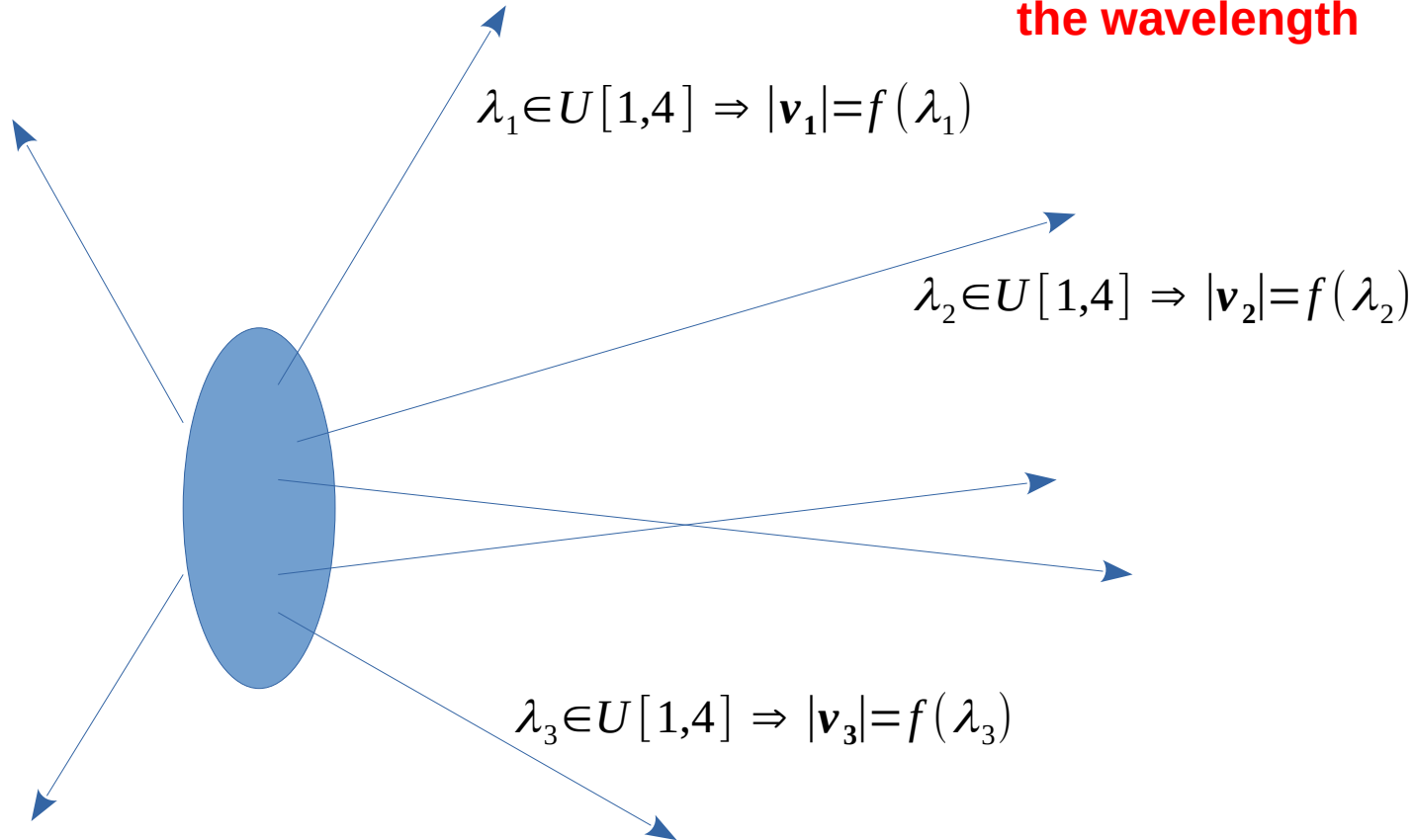
Sources: Example 1



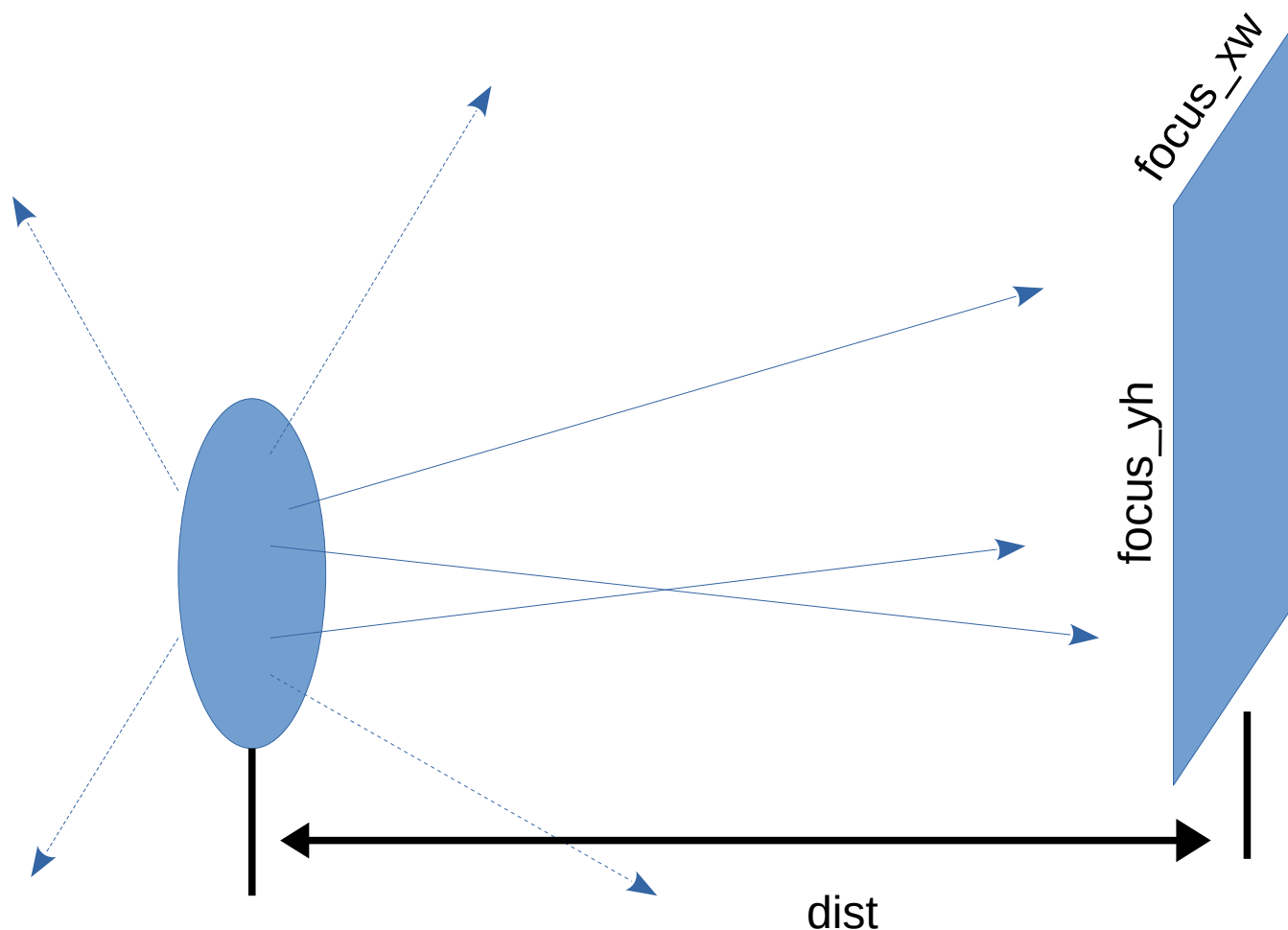
Neutron spatial coordinates are picked from a uniform distribution on a circle with radius R .

Sources: Example 1

Length of the velocity vector encodes the wavelength



Sources: Example 1

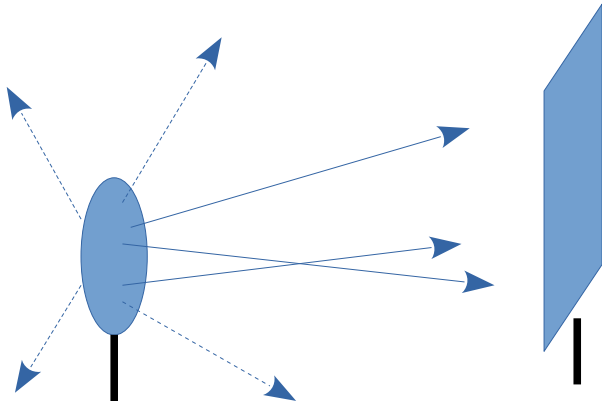


Neutron velocity vector is picked to point at a ROI.

In McStas: this is defined by the parameters: **focus_xw**, **focus_yh**, and **dist**



Sources: Example 1



TRACE

```
COMPONENT origin = Progress_bar()  
AT(0,0,0) ABSOLUTE
```

```
COMPONENT src = Source_simple(  
    radius=0.05, lambda0=2.5, dlambda=1.5,  
    focus_xw=0.1, focus_yh=0.1, dist=5)  
AT(0,0,0) RELATIVE origin
```

Monitors: in general



REALITY:

Monitors:

- Intensity probe of the beam
- Transparent to neutrons → Efficiency <1%

Detectors:

- Should detect *all* neutrons → Efficiency as high as possible

SIMULATIONS (McStas):

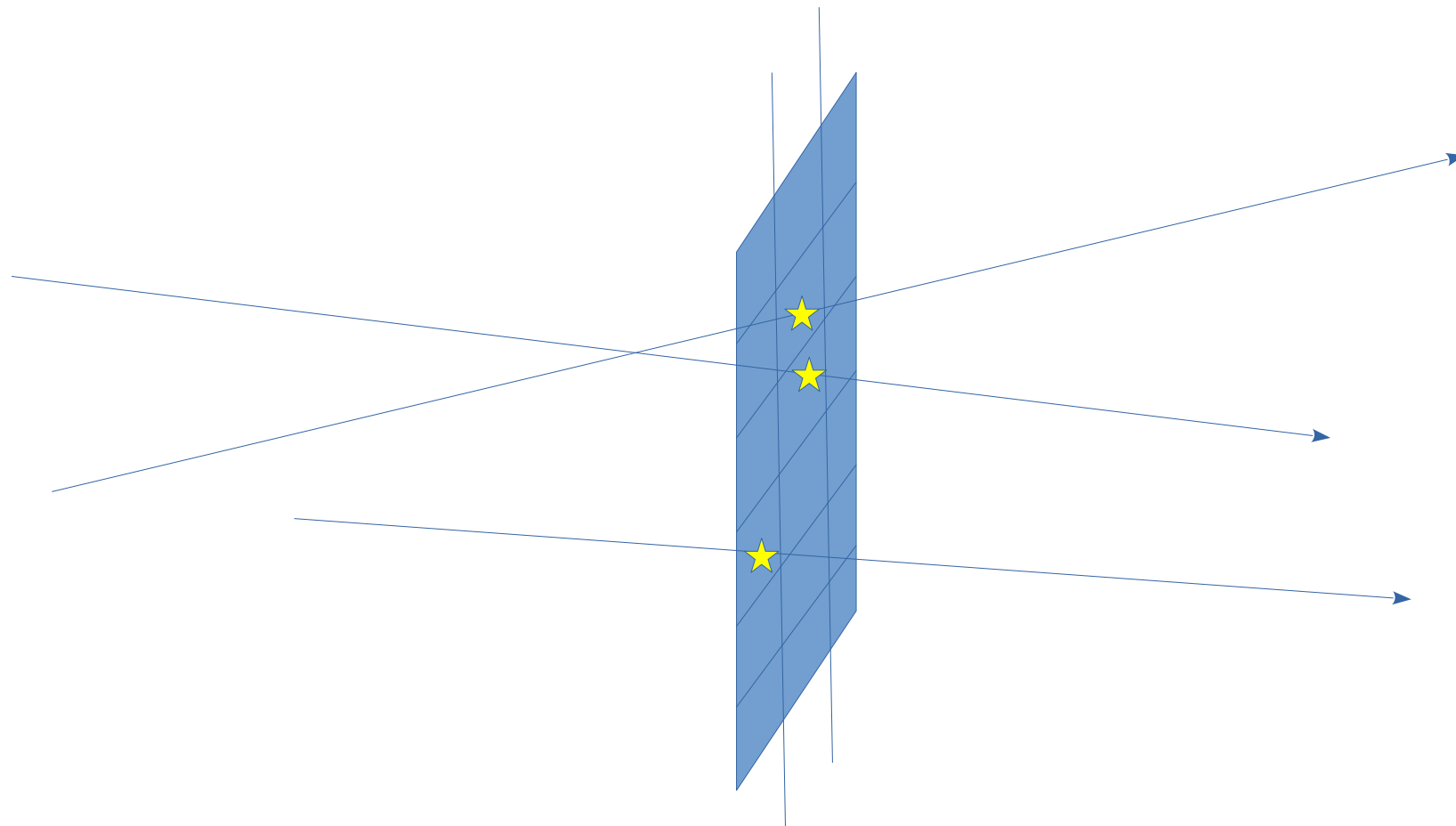
In McStas:

- We can program monitors and detectors to behave any way we like. We refer to both of those indistinguishably as 'monitors'.
- E.g. monitor with Efficiency =100% and Transparency=100%
- (With exception of PSD_Detector that models a "physical" He³ detector)

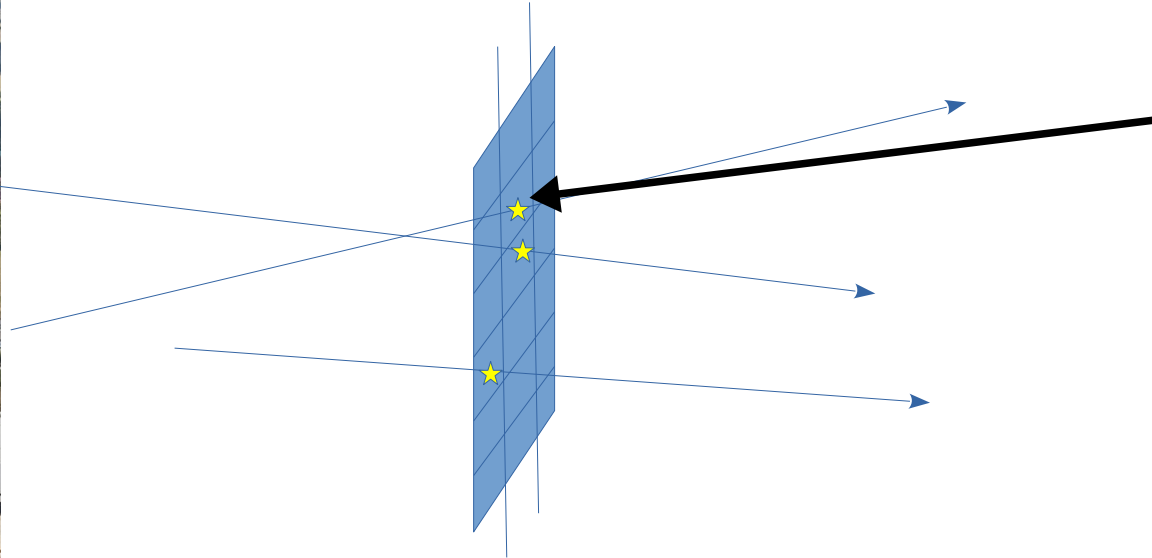


EUROPEAN
SPALLATION
SOURCE

Monitors: Example PSD_monitor



Monitors: Example PSD_monitor

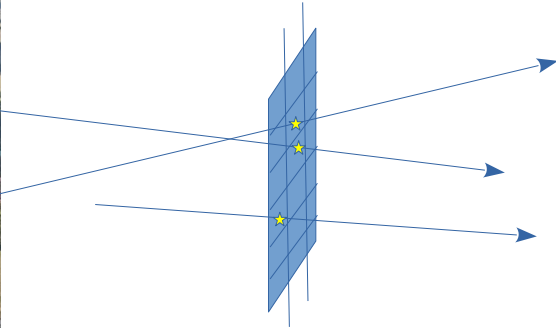


When the simulation has been completed, the detected intensity in pixel (i,j) is:

$$I(i, j) = \sum_{x_k, y_k \in \text{pixel}(i, j)} p_k; k = \text{ray number}.$$

... during simulation, the pixels are maintained as running sums.

Monitors: Example PSD_monitor and L_monitor



...

TRACE**COMPONENT** origin = Progress_bar()**AT** (0,0,0) **ABSOLUTE**
COMPONENT src = Source_simple(
 radius=0.05, lambda0=2.5, dlambd=1.5,
 focus_xw=0.1, focus_yh=0.1, dist=5)
AT (0,0,0) **RELATIVE** origin
COMPONENT psd = PSD_monitor(
 xwidth=0.2, yheight=0.2, filename="psd.dat")
AT (0,0,5) **RELATIVE** src
COMPONENT lm = L_monitor(
 xwidth=0.2, yheight=0.2, filename="lm.dat",
 Lmin=0, Lmax=8)
AT (0,0,5+0.01) **RELATIVE** src



Sources: Mathematical sources

Source_simple:

- Square or circular surface emitting neutrons from either uniform or Gaussian wavelength (or energy) distribution.
- Neutrons are directed towards a square target.

Source_div:

- Square surface emitting neutrons from either uniform or Gaussian wavelength (or energy) distribution.
- Neutrons have a divergence defined by either uniform or Gaussian distribution.



Source_simple docs

Try “**mcdoc Source_simple**”

or

(in GUI) **Help** → **mcdoc Component Reference** → (In Webpage) **Source_simple**



McStas: Source_simple - Mozilla Firefox

McStas: Source_simple x +

file:///usr/share/mcstas/2.5/sources/Source_simple.html

[[Identification](#) | [Description](#) | [Input parameters](#) | [Output parameters](#) | [Links](#)]

The Source_simple Component

A circular neutron source with flat energy spectrum and arbitrary flux

Identification

- **Site:**
- **Author:** Kim Lefmann
- **Origin:** Risoe
- **Date:** October 30, 1997

Description

The routine is a circular neutron source, which aims at a square target centered at the beam (in order to improve MC-acceptance rate). The angular divergence is then given by the dimensions of the target. The neutron energy is uniformly distributed between λ_{min} and λ_{max} or between E_{min} and E_{max} . The flux unit is specified in n/cm²/s/st/energy unit (meV or Angs).

This component replaces Source_flat, Source_flat_lambda, Source_flux and Source_flux_lambda.

Source_simple docs

Try “**mcdoc Source_simple**”

or

(in GUI) **Help** → **mcdoc Component Reference** → (In Webpage) **Source_simple**



McStas: Source_simple - Mozilla Firefox

McStas: Source_simple x +

file:///usr/share/mcstas/2.5/sources/Source_simple.html

[[Identification](#) | [Description](#) | [Input parameters](#) | [Output parameters](#) | [Links](#)]

The Source_simple Component

A circular neutron source with flat energy spectrum and arbitrary flux

Identification

- **Site:**
- **Author:** Kim Lefmann
- **Origin:** Risoe
- **Date:** October 30, 1997

Description

The routine is a circular neutron source, which aims at a square target centered at the beam (in order to improve MC-acceptance rate). The angular divergence is then given by the dimensions of the target. The neutron energy is uniformly distributed between λ_{min} and λ_{max} or between E_{min} and E_{max} . The flux unit is specified in $\text{n/cm}^2/\text{s}/\text{st}/\text{energy unit}$ (meV or Angs).

This component replaces Source_flat, Source_flat_lambda, Source_flux and Source_flux_lambda.

Click “**Input parameters**”



Source_simple docs

McStas: Source_simple - Mozilla Firefox

McStas: Source_simple x +

file:///usr/share/mcstas/2.5/sources/Source_simple.html#

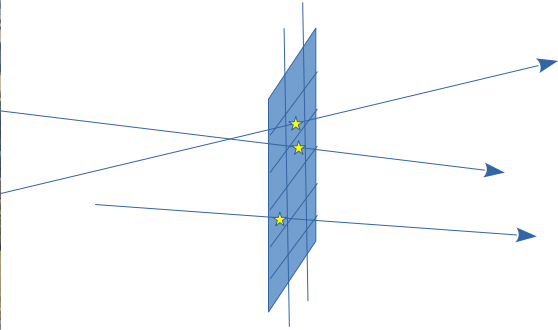
Input parameters

Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	Default
radius	m	Radius of circle in (x,y,0) plane where neutrons are generated.	0.1
yheight	m	Height of rectangle in (x,y,0) plane where neutrons are generated.	0
xwidth	m	Width of rectangle in (x,y,0) plane where neutrons are generated.	0
dist	m	Distance to target along z axis.	0
focus_xw	m	Width of target	.045
focus_yh	m	Height of target	.12
E0	meV	Mean energy of neutrons.	0
dE	meV	Energy half spread of neutrons (flat or gaussian sigma).	0
lambda0	AA	Mean wavelength of neutrons.	0
dlambda	AA	Wavelength half spread of neutrons.	0
flux	1/(s*cm**2*st*energy unit)	flux per energy unit, Angs or meV if flux=0, the source emits 1 in 4*PI whole space.	1
gauss	1	Gaussian (1) or Flat (0) energy/wavelength distribution	0
target_index	1	relative index of component to focus at, e.g. next is +1 this is used to compute 'dist' automatically.	+1



Monitors: Example PSD_monitor and L_monitor



Let's do a very simple exercise on this:

Head on over to:

Exercise 1 - Sources and Monitors on github

https://github.com/mccode-dev/Schools/tree/master/2025/ISIS_November_2025/1_Tuesday_November_4th/2_Component_basics/Exercise